$\tau - \theta$ puzzle

Parity conservation was verified in electromagnetic and strong interactions

In 1950s a particle with mass $\approx m_p/2$ was discovered

Sometimes it decayed into two pions (θ^+), sometimes into three pions (τ^+)

Pion has negative parity. So, two options:

1. Two different particles

Decay	$\theta^+ \to \pi^+ + \pi^0$	$\tau^+ \to \pi^+ + \pi^+ + \pi^-$
Parity	$+1 = (-1) \times (-1)$	$-1 = (-1) \times (-1) \times (-1)$

2. Parity is violated (?!)

Bits of history

Feynman